

BTEB NTVQF LEVEL-3: WEB DESIGN (FULL REVISION MASTER GUIDE)

Bangladesh Technical Education Board (BTEB) & ASSET Project Assessment Standard

Complete Coverage Organized Strictly by Download Folder PDF Filenames | All Questions Included A-to-Z

সূচিপত্র ও ফাইল ইনভেন্টরি (Document Inventory by PDF Name)

1. Question & Viva.pdf / NSDA Assessment.pdf

(বোর্ড প্রশ্ন: ২০টি MCQ, ১০টি Short Qs, ১০টি Viva)

3. Formation Assessment -3 (Question & Answer).pdf

(ফরমেটিভ ৩: বাংলা ও ইংরেজি ব্যাখ্যা সহ ২৯টি প্রশ্ন)

5. formative-1.pdf

(ফরমেটিভ ১: ১০টি MCQ + ২০টি Short Questions)

7. Formation Assessment -4 (Question & Answer).pdf / Suggestion-4

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9. Suggestion-1 (Web Design, L-3).pdf

(সাজেশন ১: ৩০টি MCQ + ৩০টি Short Questions)

11. Suggestion-3 (Web Design, L-3).pdf

(সাজেশন ৩: ৩২টি স্পেশাল বোর্ড প্রশ্ন)

13. Practical Job Sheet 1 to 4 Guide

(প্র্যাকটিক্যাল গাইড: Job 1 CV, Job 2 Menu/Form, Job 3 Layout, Job 4 Software Co. Site)

2. Viva Questions & Answers.pdf

(অফিসিয়াল বোর্ড ভাইভা প্রশ্নোত্তর ব্যাংক - ১০টি প্রশ্ন)

4. formative-3.pdf

(ফরমেটিভ ৩ পূর্ণাঙ্গ: ৩০টি বিস্তারিত প্রশ্ন ও কোড)

6. formative-2.pdf / Formative Assessment -2.pdf

(ফরমেটিভ ২: ২০টি MCQ + ১০টি Short Questions)

8. Formation Assessment -5 (Question & Answer).pdf / Suggestion-5

(ফরমেটিভ ৫: ৩০টি প্রশ্ন ও উত্তর)

10. Suggestion-2 (Web Design, L-3).pdf

(সাজেশন ২: কোড এডিটর ও সিএসএস মেথডস গাইড)

12. Suggestion-6 (Web Design, L-3).pdf

(সাজেশন ৬: অ্যাক্সেসিবিলিটি ও কোর টপিকস)

[PDF: Question & Viva.pdf] & [PDF: NSDA Assessment (Question and Answer).pdf]

মূল ফাইনাল বোর্ড লিখিত পরীক্ষা (২০টি MCQ + ১০টি Short Questions + ভাইভা প্রশ্ন)

অংশ ক: ২০টি বহুনির্বাচনী প্রশ্ন (MCQs with All Options & Explanations)

1. Why is it important to consider the desired outcome of all parties during negotiations?

a) To ensure fairness | b) All of the above | c) To achieve a mutual agreement | d) To avoid conflict

✓ **সঠিক উত্তর: b) All of the above**

💡 ব্যাখ্যা: আলোচনার মূল লক্ষ্য হলো উভয় পক্ষের ন্যায্যতা (fairness), পারস্পরিক সম্মতি (mutual agreement) নিশ্চিত করা এবং সংঘাত (conflict) এড়ানো।

2. Which of the following is a common health risk specific to IT workplaces?

a) Hearing loss from constant loud noise | b) Risk of drowning | c) Exposure to extreme cold | d) Repetitive Strain Injury (RSI) from prolonged keyboard and mouse use

✓ **সঠিক উত্তর: d) Repetitive Strain Injury (RSI) from prolonged keyboard and mouse use**

💡 ব্যাখ্যা: একটানা কীবোর্ড ও মাউস ব্যবহারের ফলে হাত ও আঙুলে RSI বা কার্পাল টানেল সিন্ড্রোম দেখা দেয়।

3. How GitHub used for in web development?

a) Designing graphics | b) Writing content | c) Creating animations | d) Hosting and collaborating on code

✓ **সঠিক উত্তর: d) Hosting and collaborating on code**

Example: `git push origin main` ব্যবহার করে গিটহাবে কোড হোস্ট ও শেয়ার করা হয়।

4. What is the default port for HTTP?

a) 22 | b) 80 | c) 443 | d) 3306

✓ **সঠিক উত্তর: b) 80**

Shortcut HTTP = 80 | HTTPS = 443 | SSH = 22 | MySQL = 3306

5. Web-safe colors are:

a) Colors that display consistently across browsers and devices | b) Colors only for print | c) Random colors | d) Colors that are invisible

✓ **সঠিক উত্তর: a) Colors that display consistently across browsers and devices**

💡 ব্যাখ্যা: ২১৬টি ওয়েব-সেফ কালার যেকোনো ডিভাইস ও ব্রাউজারে একই রকম দেখায়।

6. Offline tools to manage color palettes can include:

a) Terminal commands | b) Adobe Illustrator, Photoshop, or Sketch | c) Browser cache | d) HTML editors only

✓ **সঠিক উত্তর: b) Adobe Illustrator, Photoshop, or Sketch**

7. Which color is commonly used for warning signs or alerts?

a) Green | b) Blue | c) Red | d) Purple

✓ সঠিক উত্তর: c) Red

Color Psychology Red = Alert / Danger / Error | Green = Success | Blue = Trust / Info

8. Which accessibility principle focuses on readable color contrast?

✓ সঠিক উত্তর: c) Perceivable (সহজে দৃশ্যমান বা ইন্দ্রিয়গ্রাহ্য)

a) Robust | b) Understandable | c) Perceivable | d) Operable

💡 ব্যাখ্যা: WCAG-এর ৪টি মূল স্তম্ভ হলো POUR: Perceivable, Operable, Understandable, Robust.

9. Which of the following tags is used to define a table header?

a) <tr> | b) <td> | c) <th> | d) <table>

✓ সঠিক উত্তর: c) <th> (Table Heading)

Example: <th>Student Name</th> (স্বাভাবিকভাবেই বোল্ড ও সেন্টারড হয়)

10. How do you create an anchor link that points to another part of the same page?

a) Link to Section 1 | b) | c) | d)

✓ সঠিক উত্তর: a) Link to Section 1

Example: Contact Us ক্লিক করলে সরাসরি <div id="contact"> এ স্ক্রোল হবে।

11. What does the <head> section of an HTML document contain?

a) Main visible content | b) Stylesheets, scripts, and metadata | c) Only images | d) Footer information

✓ সঠিক উত্তর: b) Stylesheets, scripts, and metadata

12. How do you center a background image in CSS?

a) background-align: center; | b) background: center-image; | c) background-position: center; | d) image-align: middle;

✓ সঠিক উত্তর: c) background-position: center;

Example: `body { background-image: url('hero.jpg'); background-position: center; background-repeat: no-repeat; }`

13. How do you make a div take up the full height of the viewport?

a) height: 100vh; | b) height: 100%; | c) height: full; | d) height: 100vw;

✓ সঠিক উত্তর: a) height: 100vh;

Shortcut vh = Viewport Height (100vh = ব্রাউজার উইন্ডোর সম্পূর্ণ উচ্চতা)।

14. How do you include an external CSS file in an HTML document?

a) `<style src="style.css">` | b) `<link rel="stylesheet" href="style.css">` | c) `<script href="style.css">` | d) `<css>style.css</css>`

✓ সঠিক উত্তর: b) `<link rel="stylesheet" href="style.css">`

Example: এটি `<head>` ট্যাগের ভেতরে লিখতে হয়।

15. Which of the following software is commonly used for designing web layouts?

a) Microsoft Word | b) VLC Player | c) Figma | d) Google Chrome

✓ সঠিক উত্তর: c) Figma

16. Which of the following file formats is scalable and commonly used for web icons?

a) JPG | b) PNG | c) SVG | d) BMP

✓ সঠিক উত্তর: c) SVG (Scalable Vector Graphics)

💡 ব্যাখ্যা: SVG ভেক্টর ফাইল হওয়ায় যত বড়ই করা হোক না কেন কোয়ালিটি নষ্ট বা পিক্সেল ফেটে যায় না।

17. How can you ensure that a design is accessible on both large screens and small mobile devices?

a) Using fixed widths in pixels | b) Creating a responsive design using CSS Flexbox, Grid, and Media Queries | c) Designing only for desktop | d) Disabling mobile view

✓ সঠিক উত্তর: b) Creating a responsive design using CSS Flexbox, Grid, and Media Queries

18. What will be the output of the following code?

```
let x = 10; if (x > 5) { console.log("Yes"); } else { console.log("No"); }
```

a) No | b) undefined | c) Error | d) Yes

✓ সঠিক উত্তর: d) Yes

💡 ব্যাখ্যা: $10 > 5$ সত্য (true), তাই if ব্লক কার্যকর হয়ে "Yes" প্রিন্ট হবে।

19. What is the difference between the == and === operators in JavaScript?

a) No difference | b) == checks value only (with type conversion), === checks both value and data type | c) === checks value only | d) == is assignment

✓ সঠিক উত্তর: b) == checks value only, === checks both value and data type

Example: `5 == "5"` is true | `5 === "5"` is false (কারণ একটি number, অন্যটি string)

20. What is an open-ended question during client interviews?

a) A question answered with Yes/No | b) A question that allows the client to provide detailed explanations | c) A multiple-choice question | d) A question with no answer

✓ সঠিক উত্তর: b) A question that allows the client to provide detailed explanations

Example: "What features do you envision for this website?" (ক্লায়েন্ট বিস্তারিত বলতে পারে)।

অংশ খ: ১০টি সংক্ষিপ্ত প্রশ্ন ও উত্তর (Short Questions Q21–Q30 with Code Examples)

21. Mention one electrical safety hazard in the workplace.

উত্তর: Overloading electrical outlets with too many high-power devices or using damaged frayed cords.

বাংলা: একটি মাল্টিপ্লাগে অতিরিক্ত কম্পিউটার বা ডিভাইস সংযোগ দিলে ওভারহিটিং হয়ে আগুন লাগার ঝুঁকি তৈরি হয়।

Example: Plugging 10 desktop CPUs into a single 5A power strip.

22. What is a media query in CSS?

উত্তর: A media query is a CSS technique used to apply different styles for different device screen widths and types to ensure responsive design.

Example: `@media (max-width: 768px) { body { font-size: 14px; } }`

23. What is modular design in UI?

উত্তর: Modular design is a UI approach where the interface is broken down into small, independent, and reusable components.

বাংলা: সম্পূর্ণ লেআউটকে ছোট ছোট পুনঃব্যবহারযোগ্য ব্লকে ভাগ করা (যেমন: Navbar, Card, Footer)।

Example: Creating a reusable `<div class="product-card">` used across 10 pages.

24. What is meant by mobile-first design?

উত্তর: Mobile-first design is a strategy where websites are designed for small mobile screens first, and then scaled up for tablets and desktops using CSS media queries (min-width).

Example: Base CSS writes mobile layout; `@media (min-width: 992px)` adds desktop 3 columns.

25. What is the difference between a semantic and a non-semantic element in HTML?

উত্তর: Semantic elements clearly describe their meaning to the browser and developer (e.g., `<header>`, `<nav>`, `<footer>`). Non-semantic elements have no specific meaning and act purely as containers (e.g., `<div>`, `<span>`).

Example: Semantic: `<header>` | Non-semantic: `<div class="header">`

26. How do you select elements by class in CSS?

উত্তর: By using a period (.) followed by the class name.

Example: `.btn-primary { background-color: blue; color: white; }`

27. Highlight the first letter of a paragraph in CSS.

উত্তর: Using the `::first-letter` pseudo-element selector.

Example: `p::first-letter { font-size: 24px; font-weight: bold; color: #2563eb; }`

28. Write HTML code for a simple form with a class name.

উত্তর:

```
<form class="contact-form">
  <label>Name:</label> <input type="text" name="name" required>
  <label>Email:</label> <input type="email" name="email" required>
  <button type="submit">Send</button>
</form>
```

29. What is visual hierarchy, and why is it important in web design?

উত্তর: Visual hierarchy is the arrangement and styling of elements in order of visual importance using size, color, contrast, and spacing. It is important because it guides the user's eye directly to key information and call-to-actions.

Example: Making the main heading large (h1 32px) and secondary text smaller (p 14px).

30. What is a loop in JavaScript?

উত্তর: A loop is a control structure used to execute a block of code repeatedly as long as a specified condition evaluates to true.

Example: `for (let i = 1; i <= 5; i++) { console.log("Count: " + i); }`

[PDF: Viva Questions & Answers.pdf]

অফিসিয়াল বোর্ড ভাইভা প্রশ্নোত্তর ব্যাংক (Top 10 Board Viva Questions with Exact Answers)

Viva Q1: How can you present a PowerPoint slideshow?

উত্তর: Go to the Slide Show tab in the ribbon and click "From Beginning" (Shortcut: F5) or "From Current Slide" (Shortcut: Shift + F5). During presentation, navigate slides using arrow keys or mouse clicks.

Viva Q2: What is a formula in Excel? Give an example.

উত্তর: An expression used to perform calculations on data. It always starts with an equals sign (=). Example: =A1+B1 or =SUM(A1:A10). Formulas automate calculations and reduce manual errors.

Viva Q3: How does jQuery handle event handling?

উত্তর: jQuery simplifies event handling using built-in methods like `.click()`, `.hover()`, and `.on()`. It attaches event listeners directly to elements using concise syntax.

Example: `$("#btn").click(function() { alert("Button clicked!"); });`

Viva Q4: What are the advantages of using Bootstrap?

উত্তর: 1) Responsive Design (mobile-first), 2) Pre-built Components (navbar, modal, card), 3) 12-Column Grid System, 4) Cross-browser Compatibility, 5) Faster Development time.

Viva Q5: What are examples of popular CMS platforms?

উত্তর: 1) WordPress (most popular, easy to customize with themes/plugins), 2) Joomla (flexible and powerful), 3) Drupal (highly secure, enterprise-grade).

Viva Q6: Tell three names of JavaScript data types.

উত্তর: 1) String (e.g., "Hello"), 2) Number (e.g., 25, 3.14), 3) Boolean (true/false). Other types: Undefined, Null, Object, Symbol.

Viva Q7: Which loop is used to iterate over the properties of an object in JavaScript?

উত্তর: The `for...in` loop is used to iterate over the enumerable properties (keys) of an object.

Example: `let person = {name: "John", age: 25}; for (let key in person) { console.log(key + ": " + person[key]); }`

Viva Q8: Which HTML tag is used to define the main content of a web page?

উত্তর: The `<main>` tag. It represents the dominant and unique content of the page body, excluding headers, footers, sidebars, and navigation bars.

Example: `<main><article><h1>Article Title</h1></article></main>`

Viva Q9: How can you target an HTML element with a specific id in CSS?

উত্তর: In CSS, we use the hash (#) symbol followed by the unique id name. An ID must be unique per HTML page.

Example: `#header { background-color: #1e3a8a; color: white; }`

Viva Q10: Which CSS property is used to change the text color of an element?

উত্তর: The color property. Values can be color names (red), HEX codes (#ff0000), RGB (rgb(255,0,0)), or RGBA for transparency.

Example: `p { color: #2563eb; }`

[PDF: Formation Assessment -3 (Question & Answer).pdf]

ফরমেশন অ্যাসেসমেন্ট ৩ — বাংলা ও ইংরেজি বিশদ ব্যাখ্যা সহ ২৯টি প্রশ্ন ও উত্তর

1. How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?

Answer: Ensure accessibility by using semantic HTML, ARIA (Accessible Rich Internet Applications) attributes, keyboard navigation support, screen reader compatibility, and testing with accessibility tools (like Lighthouse or axe).

বাংলা: Semantic HTML ব্যবহার করে, ARIA অ্যাট্রিবিউট প্রয়োগ করে, কীবোর্ড নেভিগেশন নিশ্চিত করে এবং Lighthouse বা axe টুলের মাধ্যমে পরীক্ষা করে অ্যাক্সেসিবিলিটি নিশ্চিত করা হয়।

2. Which image format supports transparency and is commonly used for web graphics?

Answer: PNG (Portable Network Graphics). SVG also supports vector transparency.

3. Explain the difference between == and === operators in JavaScript.

Answer: == compares values after automatic type conversion (loose equality). === compares both value and data type without type conversion (strict equality).

Example: `5 == "5" → true` | `5 === "5" → false`

4. What are logical operators?

Answer: Logical operators are used to combine multiple conditional expressions: && (AND), || (OR), ! (NOT).

Example: `if (age >= 18 && hasNID === true) { ... }`

5. How do you use the break statement in loops and switches in JavaScript?

Answer: In loops: exits the loop immediately. In switch: stops execution from falling through after a matched case.

Example: `for (let i=1; i<=5; i++) { if (i===3) break; console.log(i); } // Prints 1, 2`

6. How do you display output in JavaScript?

Answer: 1) `console.log()` (browser console), 2) `alert()` (popup alert dialog), 3) `document.write()` (directly writes to document), 4) `element.innerHTML` (into HTML elements).

7. What are single-line and multi-line comments in JavaScript?

Answer: Single-line: `// comment` | Multi-line: `/* comment */`

8. Write some of the basic data types in JavaScript.

Answer: String, Number, Boolean, Undefined, Null, Object, Symbol (ES6), BigInt.

9. How do you check the data type of a variable in JavaScript?

Answer: Using the `typeof` operator.

Example: `typeof "Hello" // "string"` | `typeof 42 // "number"` | `typeof true // "boolean"`

10. What are arithmetic operators in JavaScript? Provide examples.

Answer: Operators used for mathematical calculations: + (addition), - (subtraction), * (multiplication), / (division), % (modulus/remainder), ** (exponentiation).

Example: $10 \% 3 = 1$ | $2^{**} 3 = 8$

11. Which keywords are used to declare a variable in JavaScript?

Answer: var (function-scoped, re-declarable), let (block-scoped, re-assignable), const (block-scoped, constant value).

12. Which operator is used for strict equality comparison in JavaScript?

Answer: Strict equality operator: === (checks both value and data type).

13. Which function is used to output data to the console in JavaScript?

Answer: console.log()

14. Which built-in method is used to convert a string to an integer in JavaScript?

Answer: parseInt() (or parseFloat() for decimal floating-point numbers).

Example: parseInt("120px") returns 120 (number).

15. What is the purpose of Tailwind CSS and how does it differ from Bootstrap?

Answer: Tailwind CSS is a utility-first CSS framework providing low-level classes (e.g., flex, pt-4, text-center) for custom styling without pre-built UI components. Bootstrap provides pre-designed components (navbar, cards, modal).

বাংলা: Tailwind হলো utility-based যেখানে ছোট ছোট ক্লাস দিয়ে সম্পূর্ণ কাস্টম ডিজাইন করা যায়। Bootstrap এ আগে থেকেই রেডিমেড কম্পোনেন্ট থাকে।

16. What is the difference between innerText and innerHTML in JavaScript?

Answer: innerText returns only visible text content and ignores HTML tags (respects CSS display:none). innerHTML returns full HTML content including child tags and parses HTML markup.

Example: el.innerText = "Hi" renders raw text; el.innerHTML renders bold text.

17. How does an interactive website differ from static and dynamic websites?

Answer: Static: Fixed content (HTML/CSS) with no updates without code edit. Dynamic: Content updates from server database (e.g., PHP/Node.js). Interactive: Real-time user interactions, form validation, animations via client-side JavaScript.

18. Which code editor do you prefer for web design and development, and why?

Answer: Visual Studio Code (VS Code). Reasons: 1) Lightweight and fast, 2) Rich extension library (Live Server, Prettier), 3) Built-in terminal and Git integration, 4) Intelligent IntelliSense code auto-completion.

19. Can you explain the basic purpose of FTP (File Transfer Protocol) software in web development?

Answer: FTP software (e.g., FileZilla) is used to upload, download, and manage website files between a local computer and a remote web hosting server.

20. How do you create an HTML login form with a submit button?

Answer:

```
<form action="login.php" method="POST">
  <label>Username:</label> <input type="text" name="username" required><br>
  <label>Password:</label> <input type="password" name="password" required><br>
  <button type="submit">Login</button>
</form>
```

21. When should I consider using SVG images?

Answer: Use SVG for logos, icons, illustrations, and UI graphics that require high resolution across retina displays and need to scale infinitely without losing quality. SVG can also be styled with CSS.

22. What is an ID selector in CSS?

Answer: An ID selector selects a single unique HTML element using the hash (#) symbol followed by the element's id attribute.

Example: #main-navbar { background: #333; }

23. How do you create functions in JavaScript?

Answer: 1) Function Declaration: `function greet() { ... }` 2) Arrow Function: `const greet = () => { ... }`

Example: `function add(a, b) { return a + b; }`

24. How do you create an ordered list in HTML?

Answer: Using the `<ol>` tag with `<li>` list item tags inside.

Example: `<ol><li>HTML</li><li>CSS</li><li>JS</li></ol>`

25. What are arrays and objects in JavaScript?

Answer: Array: An ordered list of values accessed by numeric index (let arr = [10, 20, 30]). Object: A collection of key-value pairs accessed by property name (let obj = {name: "John", age: 25}).

26. What are the three main components involved in a Client-Server setup?

Answer: 1) Client (user's browser/device), 2) Server (hosts files and database), 3) Network/Internet (communication medium via HTTP/HTTPS).

27. What is the significance of adhering to web standards in achieving cross-browser compatibility?

Answer: Adhering to W3C web standards ensures that HTML, CSS, and JS code behaves, renders, and functions identically across all modern web browsers (Chrome, Firefox, Safari, Edge).

28. What role does a database play in a CMS?

Answer: The database stores all dynamic website content, posts, pages, user accounts, comments, configuration settings, and metadata securely (e.g., MySQL in WordPress).

29. How can you create a responsive navigation bar using Bootstrap?

Answer: By using Bootstrap classes: `.navbar`, `.navbar-expand-lg`, `.navbar-toggler` (with hamburger icon), and `.navbar-collapse`.

বাংলা: স্ক্রিন ছোট হলে মেনুটি নিজে থেকেই collapse হয়ে হ্যামবার্গার আইকনে পরিণত হয়।

[PDF: formative-3.pdf]

ফরমেটিভ অ্যাসেসমেন্ট ৩ (পূর্ণাঙ্গ ৩০টি প্রশ্ন ও বিশদ কোড)

1. How do you handle conditional statements in JavaScript?

Answer: Use if, else if, and else statements for evaluating conditions, or switch statement for testing a single variable against multiple case values.

```
Example: if (score >= 80) { grade = 'A+'; } else { grade = 'Pass'; }
```

2. What is CSS? How many types of CSS? Describe with examples.

Answer: CSS stands for Cascading Style Sheets. It controls presentation, formatting, and layout. 3 Types: 1) Inline CSS (style attribute in tag), 2) Internal CSS (<style> tag in <head>), 3) External CSS (external .css file linked via <link>).

Example Inline: <p style="color:red;">Text</p>

Example Internal: <style> p { color: green; } </style>

Example External: <link rel="stylesheet" href="style.css">

3. What is the difference between JAVASCRIPT, BOOTSTRAP, JQUERY?

Answer:

Feature	JavaScript	Bootstrap	jQuery
Type	Programming Language	CSS/Frontend Framework	JavaScript Library
Main Use	Logic & Interactivity	Design, Grid & Layout	Simplifies JS DOM scripting
Dependency	None (Core)	Uses CSS + JS	Depends on JavaScript
Analogy	 Brain (Logic)	 Looks (Design)	 Shortcut (Easy syntax)

4. What is the difference between margin & padding?

Answer: Margin is the space OUTSIDE an element's border (supports 'auto' and negative values). Padding is the space INSIDE the border around the content (does NOT support negative values).

```
Example: .box { margin: 20px; padding: 15px; border: 1px solid #333; }
```

5. What are the differences between HTML Tag, Element & Attribute?

Answer: Tag: Syntax markers like <p> and </p>. Element: Complete start tag + content + end tag (e.g., <p>Hello</p>). Attribute: Extra configuration placed inside the opening tag (e.g., class="intro", src="pic.jpg").

6. What are three common features of a CMS?

Answer: 1) WYSIWYG Content Editing, 2) User Management & Roles (Admin, Editor, Author), 3) Theme & Plugin Customization support.

7. What is an IDE? Give examples.

Answer: IDE stands for Integrated Development Environment. It is a comprehensive software application providing code editor, compiler/interpreter, debugger, and terminal in a single GUI (e.g., VS Code, WebStorm, PyCharm).

8. Explain the difference between for loop, while loop, and do-while loop.

Answer: for loop: Used when number of iterations is known before. while loop: Checks condition first; runs as long as condition is true. do-while loop: Runs body AT LEAST ONCE before evaluating condition.

```
Example do-while: let i = 1; do { console.log(i); i++; } while (i <= 0); // Runs 1 time
```

9. What is CSS Universal Selector?

Answer: Represented by an asterisk (*). It selects every single HTML element on the page. Frequently used in CSS Resets.

```
Example: * { margin: 0; padding: 0; box-sizing: border-box; }
```

10. What is a rowspan and colspan in HTML tables?

Answer: rowspan specifies how many rows a cell spans vertically. colspan specifies how many columns a cell spans horizontally.

```
Example: <td rowspan="2">Spans 2 Rows</td> | <td colspan="3">Spans 3 Columns</td>
```

11. What syntax is used to include a video on a webpage?

Answer: Using the HTML5 <video> element with <source> tag and controls attribute.

```
Example:
<video width="400" controls>
  <source src="intro.mp4" type="video/mp4">
  Your browser does not support HTML video.
</video>
```

12. What are plugins in HTML / Web?

Answer: Reusable external code packages or add-ons that add extra interactive capabilities to a webpage, such as image carousels (Slick/Slider), form datepickers, charts, and video players.

13. What are the benefits of using a Headless CMS with a static website?

Answer: 1) Faster page loads (static HTML generation), 2) Greater security (no direct database exposure to users), 3) Omnichannel content delivery via APIs to web, mobile apps, and IoT devices.

14. What are the key components that make up the infrastructure of the Internet?

Answer: Web Servers, Clients, Routers, Switches, DNS (Domain Name System), IP Addresses (IPv4/IPv6), and Communication Protocols (TCP/IP, HTTP/HTTPS).

 [PDF: formative-1.pdf]

ফরমেটিভ অ্যাসেসমেন্ট ১ — ১০টি MCQ ও ২০টি শর্ট প্রশ্ন ও উত্তর

MCQ (১-১০):

1. What does CSS stand for in web design?

✓ Answer: c) Cascading Style Sheets

2. Which tag is used to create a hyperlink?

✓ Answer: c) <a> (Anchor tag)

3. Which programming language is primarily used for client-side scripting in web development?

✓ Answer: c) JavaScript

4. Which of the following is NOT a commonly used web design file format for images?

✓ Answer: b) .mp3 (Audio file format, not image)

5. What is the purpose of the "alt" attribute in HTML image tags?

✓ Answer: c) It provides alternative text for accessibility and when images fail to load

6. How can you include CSS in an HTML document?

✓ Answer: d) All of the above (Inline style attribute, <style> tag, and external <link> tag)

7. Which CSS property is used to control spacing between lines of text?

✓ Answer: d) line-height

Example: `p { line-height: 1.6; }`

8. What does the "float" property in CSS do?

✓ Answer: b) It removes an element from normal document flow and places it to the left or right

9. What does the "z-index" property in CSS control?

✓ Answer: b) It controls the stacking order of positioned elements (which sits on top)

Example: `.modal { position: fixed; z-index: 1050; }`

10. Which HTML tag is used to define a table row?

✓ Answer: a) <tr> (Table Row)

শর্ট প্রশ্ন (১১-৩০):

11. Is it possible to SEO a WordPress website to show it on Google's first page?

Answer: Yes. WordPress is naturally SEO-friendly. By optimizing high-quality content, keywords, site speed, mobile responsiveness, and utilizing SEO plugins like Yoast or Rank Math, a website can rank on Google's first page.

12. Differentiate between WordPress.com and WordPress.org.

Answer: WordPress.com is a hosted platform with managed hosting but limited theme/plugin customization. WordPress.org provides free open-source software for self-hosting with 100% full control and unlimited customization.

13. Why do we use Jumbotron in Bootstrap?

Answer: Jumbotron creates a large, prominent showcase container to highlight key headlines, marketing slogans, or call-to-action buttons on landing pages.

Example: `<div class="jumbotron"><h1>Welcome!</h1></div>`

14. What is the Bootstrap Grid System?

Answer: A mobile-first, 12-column responsive layout system that uses containers, rows, and columns to organize content across screen sizes (col-sm, col-md, col-lg).

Example: `<div class="row"><div class="col-md-6">Half</div><div class="col-md-6">Half</div></div>`

15. What are contextual classes in Bootstrap tables?

Answer: Classes used to add color-coded meaning to rows or cells: .table-primary, .table-success (green), .table-warning (yellow), .table-danger (red), .table-info (light blue).

16. What is a pseudo-class in CSS?

Answer: A keyword added to a selector specifying a special state of the element. Examples: :hover, :active, :focus, :visited, :nth-child().

17. How do you write a pseudo-element in CSS?

Answer: Pseudo-elements start with a double colon (::). They style a specific part of an element. Examples: ::before, ::after, ::first-line, ::first-letter.

18. What is a definition list in HTML?

Answer: A list of terms and their corresponding descriptions using <dl> (list), <dt> (definition term), and <dd> (definition description).

Example: `<dl><dt>HTML</dt><dd>HyperText Markup Language</dd></dl>`

19. What is Hoisting in JavaScript? Explain with example.

Answer: Hoisting is JavaScript's default behavior where variable and function declarations are moved to the top of their scope before code execution. `var` is hoisted and initialized with `undefined`; `let`/`const` are in the Temporal Dead Zone.

```
Example: console.log(x); // outputs: undefined (no crash) | var x = 10;
```

20. What are the key components of Bootstrap?

Answer: 1) CSS resets & typography, 2) 12-column Grid System, 3) Pre-built UI components (cards, badges, navbars), 4) Interactive JavaScript plugins (modal, dropdown, carousel).

 [PDF: formative-2.pdf] & [PDF: Formative Assessment -2 (Question & Answer).pdf]

ফরমেটিভ অ্যাসেসমেন্ট ২ — ২০টি MCQ ও ১০টি শর্ট প্রশ্ন ও উত্তর

MCQ (১-২০):

1. What is the ternary operator in JavaScript?

✓ Answer: b) `?: (condition ? trueVal : falseVal)`

Example: `let status = age >= 18 ? "Adult" : "Minor";`

2. How should follow-up actions be determined after negotiations?

✓ Answer: d) They should be agreed upon by all parties

3. Which is a corrective action an IT staff can take to mitigate workplace risk?

✓ Answer: d) Securing a loose carpet tile in their cubicle to prevent tripping

4. Color theory in web design helps to:

✓ Answer: c) Create visually appealing and functional designs

5. Which element can improve visual hierarchy besides typography?

✓ Answer: b) Shapes and imagery

6. What is the main role of responsive grids (CSS Flexbox/Grid)?

✓ Answer: b) Structure content to adapt fluidly to different screen sizes

7. A color-style guide helps designers by:

✓ Answer: d) Documenting brand colors, combinations, and accessibility rules for consistent application

8. Which color is commonly associated with trust and professionalism in branding?

✓ Answer: a) Blue

9. Online tools to generate color palettes include:

✓ Answer: c) Dedicated color palette generators (e.g., Adobe Color, Coolors)

10. The `<iframe>` tag is used to:

✓ Answer: b) Embed another webpage or document inside the current page

Example: `<iframe src="https://maps.google.com" width="300" height="200"></iframe>`

11. Which attribute specifies the natural language of the HTML document?

✓ Answer: b) `lang` (e.g., `<html lang="en">`)

12. How do you prevent a background from repeating in CSS?

✓ Answer: a) background-repeat: no-repeat;

13. How do you apply a box shadow in CSS?

✓ Answer: b) box-shadow: 2px 3px 5px #888888;

14. What is the primary purpose of identifying the color palette in a design?

✓ Answer: c) To ensure brand consistency and provide a harmonious visual experience

15. In a well-structured web project, where should you store fonts?

✓ Answer: b) /assets/fonts

16. Which technique is most commonly used to make images load faster on websites?

✓ Answer: b) Use smaller file formats (WebP/SVG) and compression

17. What is the result of true && false in JavaScript?

✓ Answer: d) false

18. What is the purpose of the switch statement?

✓ Answer: b) To handle multiple conditions based on a single expression value

শর্ট প্রশ্ন (১৯-২৬):

19. Why is proper lighting important in IT workspaces?

Answer: Proper lighting reduces eye strain, eliminates glare on computer screens, improves focus, and prevents headaches, thereby increasing overall productivity.

20. How can a negotiator use 'observation skills' during negotiations?

Answer: By observing the opposing party's body language, eye contact, facial expressions, and vocal tone to detect hesitation, confidence, or flexibility, allowing them to adjust strategy.

21. Which colors are formed as secondary colors in the RGB model?

Answer: Cyan (Green + Blue), Magenta (Red + Blue), and Yellow (Red + Green). CMY are secondary in RGB.

22. How to change background color smoothly on hover in CSS?

Answer: Use the transition property together with the :hover pseudo-class.

Example:

```
button { background-color: #2563eb; transition: background-color 0.3s ease; }  
button:hover { background-color: #1d4ed8; }
```

23. What unit of measurement is commonly used in web design to define margins and spacing?

Answer: px (pixels) for precise, absolute sizing; rem and em for scalable, responsive spacing.

[PDF: Formation Assessment -4 (Question & Answer).pdf] & [PDF: Suggestion-4 (Web Design, L-3).pdf]

ফরমেটিভ অ্যাসেসমেন্ট ৪ — ডিবাগিং, বুটস্ট্রাপ, ডম/বম ও জেএসন প্রশ্নোত্তর

1. What is debugging in JavaScript?

Answer: Debugging is the process of identifying, tracing, and fixing bugs, errors, or unexpected behavior in code using tools like Chrome DevTools (Console, Breakpoints) or `console.log()`.

2. What is the difference between a Post and a Page in WordPress?

Answer: Post: Time-sensitive, blog entries organized by categories, tags, and published dates. Page: Static, timeless content that rarely changes (e.g., "About Us", "Contact", "Privacy Policy").

3. What is a Container in Bootstrap and what is it used for?

Answer: A container wraps site contents, centers it horizontally, and provides padding. Two types: `.container` (fixed responsive width) and `.container-fluid` (100% full width across all screen sizes).

4. Write down the HTML boilerplate structure.

Answer:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document Title</title>
</head>
<body>
</body>
</html>
```

5. Why do we use the CSS box-sizing property?

Answer: `box-sizing: border-box` includes padding and border within the element's total declared width and height, preventing layout breakage and unwanted horizontal scrolling.

Example: `* { box-sizing: border-box; }`

6. What is the difference between DOM and BOM?

Answer: DOM (Document Object Model): Focuses on the web page document and HTML elements (`document.getElementById`). BOM (Browser Object Model): Focuses on the browser window itself (window, history, navigator, location).

Example DOM: `document.title = "Home";` | Example BOM: `window.location.reload();`

7. Describe different types of HTML elements (Block-level vs Inline).

Answer: Block-level: Starts on a new line and occupies the entire width of its container (e.g., <div>, <p>, <h1>, <form>). Inline: Only takes up as much width as its content and does not start on a new line (e.g., , <a>, ,).

8. Create a JSON Array that contains four students' name, email, mobile & address.

Answer:

```
let students = [  
  { "name": "Arif", "email": "arif@gmail.com", "mobile": "01313363836", "address": "Dhaka" },  
  { "name": "Rahim", "email": "rahim@gmail.com", "mobile": "01711223344", "address": "Chattogram" },  
  { "name": "Karim", "email": "karim@gmail.com", "mobile": "01811556677", "address": "Sylhet" },  
  { "name": "Jamal", "email": "jamal@gmail.com", "mobile": "01911889900", "address": "Rajshahi" }  
];
```

9. How do you apply multiple CSS classes to a single HTML element?

Answer: By listing class names separated by a single space inside the class attribute.

Example: <button class="btn btn-primary active">Click Me</button>

 [PDF: Formation Assessment -5 (Question & Answer).pdf] & [PDF: Suggestion-5 (Web Design, L-3).pdf]

ফরমেটিভ অ্যাসেসমেন্ট ৫ — সিএসএস টাইপস, লুপস, বুটস্ট্রাপ টেবিল ও হেডলেস সিএমএস

1. How do you handle conditional statements in JavaScript?

Answer: Use if, else if, else to branch code logic based on Boolean conditions. Use switch statement when comparing a single expression against multiple fixed values.

2. How does an interactive website differ from static and dynamic websites?

Answer: Static: Fixed pre-written HTML content. Dynamic: Server-rendered content from database (PHP/Node). Interactive: Engages user in real-time with instant client-side updates (form validation, filters, animations) via JavaScript.

3. How do you use the break statement in loops and switches in JavaScript?

Answer: In loops: Terminates loop execution immediately when condition is met. In switch: Exits the switch block so subsequent cases are not run.

4. Describe the 3 types of CSS with pros and cons.

Answer: 1) Inline: Fast to apply single-element style, but poor maintainability. 2) Internal: Styles entire single page within <head>, but cannot be reused across other pages. 3) External: Separate .css file linked via <link>; best industry practice, highly reusable and cached by browsers.

5. What is the difference between for loop, while loop, and do-while loop?

Answer: for loop: Ideal when exact repetition count is known. while loop: Tests condition before each iteration. do-while loop: Executes at least once before checking condition at the end.

6. What are the benefits of using a Headless CMS with a static website?

Answer: 1) Blazing fast website speed (served via CDN), 2) Enhanced security (no database exposed on frontend), 3) Content can be delivered via REST/GraphQL API to web, iOS, Android simultaneously.

7. When should I consider using SVG images?

Answer: When rendering logos, icons, vector shapes, or charts that require infinite scalability across all display resolutions without quality degradation.

[PDF: Suggestion-3 (Web Design, L-3).pdf]

সাজেশন ৩ — ৩২টি বিশেষ শর্ট বোর্ড প্রশ্ন ও উত্তর (Special Short Questions Bank)

1. How does an FTP client work?

Answer: Connects to a remote web server on Port 21 using FTP/SFTP credentials, enabling the developer to upload, download, and rename website files on the live host.

2. How can you create a carousel of images using Bootstrap?

Answer: Using classes: .carousel, .carousel-inner, .carousel-item, and data-bs-ride="carousel" with next/prev control buttons.

3. How can Bootstrap be integrated into a project?

Answer: 1) Via CDN (Content Delivery Network) links in HTML <head> and before </body> (fastest method), or 2) By downloading compiled CSS/JS files and linking locally.

4. What is AJAX in jQuery?

Answer: Asynchronous JavaScript and XML. It allows sending and retrieving data from a web server asynchronously in the background without reloading the whole web page.

```
Example: $.ajax({ url: "api/data.json", success: function(result) { $("#box").html(result); } });
```

5. How does jQuery simplify DOM manipulation?

Answer: jQuery uses short CSS-like selector syntax \$(selector).action() which reduces multi-line vanilla JS into a single readable line.

```
Example: $("#msg").text("Updated"); instead of document.getElementById("msg").innerText = "Updated";
```

6. What is a function in Excel? Write an example.

Answer: A built-in predefined formula in Microsoft Excel used to perform automated mathematical, statistical, or logical calculations.

```
Example: =SUM(A1:A10) calculates the total sum of cells A1 through A10; =AVERAGE(B1:B5).
```

7. What is the purpose of slide transitions in PowerPoint?

Answer: Visual animated motion effects that occur when navigating from one slide to the next during a presentation, improving audience engagement.

8. How can you spell check a document in Microsoft Word?

Answer: Go to the Review tab in the ribbon and click "Spelling & Grammar" (or press the F7 shortcut key).

9. What are the fundamental steps of Image Processing?

Answer: 1) Image Acquisition (capturing), 2) Image Enhancement (improving quality), 3) Image Segmentation (partitioning into regions), 4) Feature Extraction & Recognition.

10. What is the purpose of the :nth-child() pseudo-class in CSS3?

Answer: Selects elements based on their position index among sibling elements. Useful for zebra-stripping tables or picking specific list items.

```
Example: tr:nth-child(even) { background-color: #f2f2f2; } | li:nth-child(2) { color: red; }
```

11. What is the default positioning scheme in CSS?

Answer: static. The element renders in order according to the normal page document flow; top/right/bottom/left properties have no effect.

12. Which property is used to add shadows to text in CSS?

Answer: text-shadow: x-offset y-offset blur-radius color;

```
Example: h1 { text-shadow: 2px 2px 4px #888888; }
```

13. What are the basic SVG tags?

Answer: <svg> (root container), <circle>, <rect>, <line>, <path>, <polygon>, <text>.

```
Example: <svg width="50" height="50"><circle cx="25" cy="25" r="20" fill="green"/></svg>
```

1. How many types of layout are available in Bootstrap?

Answer: Two major layouts: 1) Fluid Layout (100% full width across viewport, uses .container-fluid), 2) Fixed Layout (fixed responsive container with max-width per breakpoint, uses .container).

2. What are the timing methods in JavaScript and why do we use them?

Answer: 1) setTimeout(func, delay): Executes a function once after a specified time delay (in ms). 2) setInterval(func, interval): Executes a function repeatedly at specified intervals. 3) clearTimeout() and clearInterval() cancel timers.

Example:

```
let timer = setTimeout(() => alert("Welcome!"), 3000);  
let counter = setInterval(() => updateClock(), 1000);
```

3. What is JSON and give an example?

Answer: JSON (JavaScript Object Notation) is a lightweight, human-readable, language-independent data-interchange format. Used to transmit data between server and client web application.

Example: let userJSON = '{"name": "Rubel", "age": 28, "city": "Dhaka"}';

4. Describe different types of repeated statements (Loops) in JavaScript.

Answer: 1) for loop (standard index iteration), 2) while loop (condition checked before execution), 3) do...while loop (executes once before condition check), 4) for...in loop (iterates over object keys), 5) for...of loop (iterates over iterable values like arrays).

5. Create a registration form using only HTML.

Answer:

```
<form action="register.php" method="POST">  
  <label>User Name:</label><br>  
  <input type="text" name="name" required><br><br>  
  <label>Email:</label><br>  
  <input type="email" name="email" required><br><br>  
  <label>Mobile:</label><br>  
  <input type="tel" name="mobile" required><br><br>  
  <label>Address:</label><br>  
  <textarea name="address" rows="3"></textarea><br><br>  
  <button type="submit">Register</button>  
</form>
```

6. What is the difference between JavaScript and Java?

Answer: Java is an OOP compiled backend language running on JVM (Java Virtual Machine). JavaScript is an interpreted, dynamic scripting language primarily executed inside web browsers for frontend interactivity. They are completely unrelated languages.

7. Popular Code Editors Classification

Answer:

Category	Editor Names	Key Features
Modern Lightweight	VS Code, Sublime Text, Atom	Fast, rich plugin ecosystem, Git integrated
Full IDE	IntelliJ IDEA, WebStorm, PyCharm	Deep refactoring, enterprise tools, heavyweight
Online Cloud Editors	CodePen, Replit, JSFiddle	Instant browser preview, zero installation required

 [PDF: Suggestion-6 (Web Design, L-3).pdf]

সাজেশন ৬ — অ্যাক্সেসিবিলিটি, এআরআইএ রোলস ও কোর কনসেপ্টস

1. How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?

Answer: 1) ARIA (Accessible Rich Internet Applications) attributes (e.g., aria-label, aria-expanded, role="dialog"), 2) Semantic HTML elements, 3) Full keyboard focus and navigation support (tabindex), 4) Ensuring screen readers announce dynamic DOM updates using aria-live.

2. Which image format supports transparency and is commonly used for web graphics?

Answer: PNG (Portable Network Graphics) and SVG (Scalable Vector Graphics).

3. Explain the difference between == and === in JavaScript.

Answer: == compares values after type conversion. === compares both value and data type without type conversion.

Example: `0 == false` is true | `0 === false` is false

[Practical Assessment Guide: Job Sheets 1, 2, 3, 4]

প্র্যাকটিক্যাল পরীক্ষা গাইডলাইন (২.৫ ঘণ্টা সময়কাল) — শতভাগ নম্বর পাওয়ার প্রস্তুতি

Job Sheet 1: Curriculum Vitae (CV) Formatting

কাজ ও অবস্থান: MS Word অথবা HTML/CSS দিয়ে প্রফেশনাল CV তৈরি করা।

✦ অলরেডি তৈরি করা ফাইল: d:\Web Design\cv.html এবং Desktop\CV\index.html

পরীক্ষায় যা যা লাগবে: নাম, যোগাযোগ, প্রফেশনাল সামারি, শিক্ষাগত যোগ্যতা (Table), কারিগরি স্কিল ও প্রজেক্ট লিস্ট।

Job Sheet 2: Restaurant Menu / Form Development

কাজ: খাদ্য তালিকা (Menu Items) গ্রিড বা টেবিল আকারে সাজানো এবং টেবিল বুকিং বা অর্ডার ফর্ম ডিজাইন করা।

Required Elements: Food categories, pricing tables, image cards, reservation form with date/time pickers.

Job Sheet 3: Responsive Web Page with Bootstrap

কাজ: রেসপনসিভ ব্যানার, ন্যাভবার এবং ৩-কলাম বিশিষ্ট সার্ভিস কার্ড তৈরি করা যা মোবাইলে ১ কলাম হবে।

Bootstrap Grid Classes: .container > .row > .col-lg-4 .col-md-6 .col-12

Job Sheet 4: Software Development Company Website (6 Pages)

কাজ: একটি পূর্ণাঙ্গ ৬ পেজের সফটওয়্যার কোম্পানির রেসপনসিভ ওয়েবসাইট তৈরি করা।

☀ সম্পূর্ণ প্রজেক্ট প্রস্তুত আছে: d:\Web Design\ ফোল্ডারে সম্পূর্ণ ৬টি পেজ তৈরি করা রয়েছে:

- 1) index.html (Home Page with Hero & Services)
- 2) about.html (Company About Us & Team)
- 3) services.html (Detailed IT Services)
- 4) projects.html (Portfolio Showcase)
- 5) project-detail.html (Single Case Study)
- 6) contact.html (Interactive Contact Form & Validation)
- 7) css/style.css (Custom Styling & Media Queries)